Name: Tyler Coutermarsh
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Course: Python 100
Assignment: 07
[Github Link]

Classes and Objects

Introduction

In this assignment, I added on from the previous weeks program but this time introducing data classes. This is utilized to be able to easily access the student's first and last name and course name data within the script. Like accessing the functions in their own classes from last week for simpler accessibility.

Data Classes

In the student class, I start off by creating the constructor of the student's first and last name. This creates the object for the student first and last name to be appended to the student list. The getter is then created to basically create a function that allows for easy access to later call the student first and last name in the script. The setter is created to write the data from the object.

```
class Person:  1 usage

    # TODO Add first_name and last_name properties to the constructor
    def __init__(self, first_name: str = "", last_name: str = ""):
        self.first_name = first_name
        self.last_name = last_name

    # TODO Create a getter and setter for the first_name property
    @property  4 usages (2 dynamic)
    def first_name(self):
        return self.__first_name.title()

    @first_name.setter  3 usages (2 dynamic)
    def first_name(self, value: str):
        if value.isalpha() or value == "":
            self.__first_name = value
        else:
            raise ValueError("The first name should not contain numbers.")

    # TODO Create a getter and setter for the last_name property
    @property  4 usages (2 dynamic)
    def last_name(self):
        return self.__last_name.title()

    @last_name.setter  3 usages (2 dynamic)
    def last_name(self, value: str):
        if value.isalpha() or value == "":
            self.__last_name = value
        else:
            raise ValueError("The last name should not contain numbers.")

    # TODO Override the __str__() method to return Person data
    def __str__(self):
        return f"{self.first_name},{self.last_name}"
```

The second class called Student, uses the previous Person class by basically adding the course name to the object. It also then has a getter and setter created for the course name.

```
# TODO Create a Student class the inherits from the Person class
class Student(Person):  2 usages

    # TODO call to the Person constructor and pass it the first_name and last_name data
    def __init__(self, first_name: str = "", last_name: str = "", course_name: str = ""):
        super().__init__(first_name=first_name, last_name=last_name)

        # TODO add a assignment to the course_name property using the course_name parameter
        self.course_name = course_name
```

Summary

This assignment further practices functions and classes to make the script structured better. It allows for more organization, as well as easier accessibility to not just functions but also data.